

OPTICAL FENCE

Using WeMos D1 Mini With Updates On Google Sheet

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Vigilance of any room, location, or enclosure is always challenging, especially when it cannot be physically locked. A real-life scenario requiring the tracking of intrusions in a monitored space, inspired the creation of this cost-effective solution to log movement onto a Google Sheet using an optical fence. This concept can be extended to any monitored area like those under CCTV surveillance, where constant attention to the screen is unnecessary.

The Google Sheet tracks the exact date and time of intrusions, enabling to pinpoint specific

moments and verify access through CCTV footage. Other potential applications include monitoring corridors, examination rooms, or open pathways where movement needs to be logged. The goal is not just to count persons but to track the timing of every movement. Additional buzzers or light indicators can enhance the system's functionality.

Using a simple laser light sensor setup with the WeMos D1 Mini, this concept comes to life. A 10-second timer updates the Google Sheet every 10 seconds to record inactivity, while activity prompts an

immediate update. A new 10-second timer is then initiated after each detection. Fig. 1 illustrates the proposed setup for the observation zone, with the WeMos D1 Mini and laser light assembly. The required components (bill of materials) are listed in the table.

Features and technical specifications of the laser module used in the device

The laser module used in the device is RoHS (restriction of hazardous substances) compliant and cost-effective. Its wavelength ranges from 635nm to 660nm, with a rise and

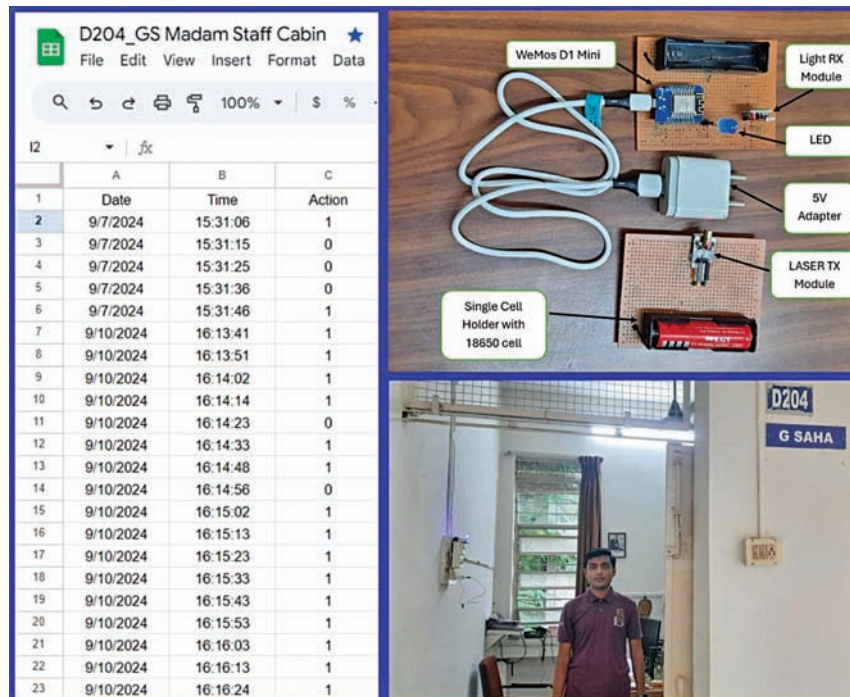


Fig. 1: Authors' proposed setup

BILL OF MATERIALS	
Components	Quantity
WeMos D1 mini (MOD1)	1
Laser transmitter	1
Optical receiver	1
Jumper wires	3
18650 rechargeable batteries (3.7V)	2
5V USB cable with adaptor	1

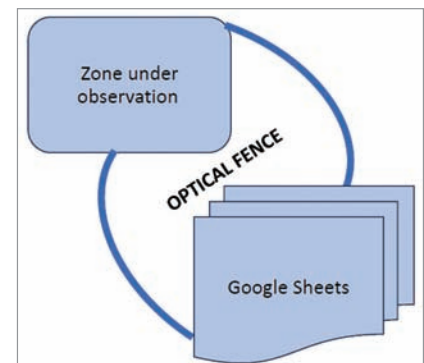


Fig. 2: Block diagram representation of the entire setup